

Assessing Generalized Anxiety Disorder in Elderly People Using the GAD-7 and GAD-2 Scales: Results of a Validation Study

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Objective: *The aim of this study was to evaluate the validity of the seven-item Generalized Anxiety Disorder scale (GAD-7) and its two core items (GAD-2) for detecting GAD in elderly people. Methods:* A criterion-standard study was performed between May and December of 2010 on a general elderly population living at home. A subsample of 438 elderly persons (ages 58–82) of the large population-based German ESTHER study was included in the study. The GAD-7 was administered to participants as part of a home visit. A telephone-administered structured clinical interview was subsequently conducted by a blinded interviewer. The structured clinical (SCID) interview diagnosis of GAD constituted the criterion standard to determine sensitivity and specificity of the GAD-7 and the GAD-2 scales. **Results:** *Twenty-seven participants met the Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition criteria for current GAD according to the SCID interview (6.2%; 95% confidence interval [CI]: 3.9%–8.2%). For the GAD-7, a cut point of five or greater appeared to be optimal for detecting GAD. At this cut point the sensitivity of the GAD-7 was 0.63 and the specificity was 0.9. Correspondingly, the optimal cut point for the GAD-2 was two or greater with a sensitivity of 0.67 and a specificity of 0.90. The areas under the curve were 0.88 (95% CI: 0.83–0.93) for the GAD-7 and 0.87 (95% CI: 0.80–0.94) for the GAD-2. The increased scores on both GAD scales were strongly associated with mental health related quality of life ($p < 0.0001$). Conclusion:* Our results establish the validity of both the GAD-7 and the GAD-2 in elderly persons. Results of this study show that the recommended cut points of the GAD-7 and the GAD-2 for detecting GAD should be lowered for the elderly general population. (Am J Geriatr Psychiatry 2014; 22:1029–1038)

Key Words: Generalized anxiety disorder, GAD-7 scale, validation, elderly

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INTRODUCTION

Generalized anxiety disorder (GAD) is one of the most prevalent mental disorders in the elderly population, with estimates ranging between 2.8% and 7.3%.^{1,2} The core symptoms of GAD are excessive and uncontrollable worries over a period of at least 6 months.^{3,4} Further criteria include autonomic arousal (e.g., palpitations, trembling), sleep disturbances, and difficulties concentrating. GAD in older people is associated with disability, increased healthcare use, and impaired quality of life.^{2,5} In addition, in older people with GAD, a comorbid depressive disorder is frequently diagnosed.^{6,7}

Unfortunately, GAD in elderly people often goes undetected⁸ because diagnosing GAD in the elderly can be difficult. Somatic diagnostic criteria, such as being easily fatigued, difficulty in concentrating, or sleep disturbances, do not easily apply to the elderly population because such symptoms are also common in the elderly who do not suffer from GAD. In addition, elderly people may tend to conceal their mental impairment and complain less about psychological burdens.⁹ Furthermore, the diagnostic process is complicated by the paucity of valid instruments to screen for (or detect) GAD in elderly people.¹⁰ Results regarding the Generalized Anxiety Disorder Severity Scale (GADSS) for instance, are inconsistent. Weiss et al.¹¹ showed that the GADSS had poor diagnostic accuracy in a sample of older primary care patients. In contrast, Andreescu et al.¹² found that the GADSS showed a good convergent and discriminant validity in elderly people. Wetherell et al.¹³ reported that the Hospital Anxiety and Depression Scale, subscale Anxiety, showed a high sensitivity and specificity for detecting older people with GAD. However, this study was conducted in only 68 elderly primary care patients who were preselected by GAD screening questions; therefore, no inferences regarding the general population can be drawn.

The seven-item Generalized Anxiety Disorder scale (GAD-7) was developed to identify cases of GAD and to assess symptom severity; the seven items are specifically linked to the *Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition* (DSM-IV) criteria. The GAD-7 was validated in a large sample of primary care patients and proved to have good psychometric properties.¹⁴ A GAD-7 score of 10 or greater was recommended as the cut point for iden-

tifying persons with GAD. The first two items of the GAD-7 reflect the core diagnostic criteria of GAD according to the DSM-IV. These two items constitute the ultrabrief version of the GAD-7, the so-called GAD-2.¹⁵ The recommended cut point for the GAD-2 is three or greater.¹⁶ The German versions of the GAD-7 and GAD-2 also proved to be reliable and valid instruments for screening for GAD.^{16,17}

However, to date no study has been conducted to investigate the validity of the GAD scales in a representative sample of older people. In this regard, it is important to determine the answer to one important question: Are the recommended cut points of the GAD-7 and the GAD-2 for younger and middle-aged adults the same as those for older adults? Because it is known that older people in general are less inclined to complain about persistent worries or negative feelings, it could be possible that the threshold for detecting GAD in elderly people should be lowered.

The aim of the present study was to examine the operating characteristics of the GAD-7 and GAD-2 scales in participants ages 58–82 years to (1) investigate the criterion and construct validity of the two GAD scales regarding their use for elderly people and (2) determine reasonable cut-offs for identifying elderly patients with GAD.

METHODS

Study Design and Participants

The study was part of the Epidemiologische Studie zu Chancen der Verhuetung, Frueherkennung und optimierten Therapie chronischer Erkrankungen (ESTHER) study and was approved by the ethics committee of the Medical University of Heidelberg and the Medical Association of the state of Saarland. The ESTHER study is a population-based cohort study in Germany. Between July 2000 and December 2002, in the federal state of Saarland, 9,949 participants aged 50–74 were recruited at baseline.¹⁸ At the beginning of the 8-year follow-up of the ESTHER study, 8,770 participants were eligible. Of these, 505 participants were not able to complete a standardized questionnaire, leaving a balance of 8,265 participants.

All in all, between 2008 and 2010, 6,063 elderly people participated in the third 8-year follow-up. All participants of the 8-year follow-up were asked if

they would participate in a longer home visit that would be conducted for personal interviews and a geriatric assessment. Of the 6,063 ESTHER participants, 3,124 agreed to be visited at home. The home visits served as a comprehensive assessment regarding functional status as well as medical, pharmacologic, socioeconomic, and psychosocial aspects of their life. The participants of the present study constitute a subsample of these home visit participants of the ESTHER study. The present study took place between May and December of 2010; the only inclusion criterion was participation in a home visit of the ESTHER study. Thus, 509 elderly ESTHER participants, visited consecutively, were asked to participate in the present study. During the home visits, the 509 eligible persons completed the GAD-7 in the presence of a study doctor.

Of these 509 eligible elderly persons, 45 (8.8%) could not be contacted by telephone after the home visit and 23 (4.5%) refused to participate in the telephone interview. Three persons were excluded because of missing data in the questionnaires. Ultimately, 438 participants (86.1%) agreed to the telephone interview and were included in the present study. No differences were observed between study participants and persons excluded from the study ($N = 68$) with regard to age, gender, education, marital status, and mean GAD-7 scores.

Measures

At the 8-year follow-up of the ESTHER study, all participants completed a self-report questionnaire that included demographic characteristics and the Short-Form General Health Survey (SF-12) to measure health-related quality of life. Items of the SF-12 are weighted and totaled to provide both physical component scores (PCS) and mental component scores (MCS) ranging from 0 to 100. A higher score in the respective summary scales indicates a higher quality of life. The questionnaire shows good psychometric criteria.¹⁹

In addition, during the home visits, the participants of the study were comprehensively assessed by the study doctors. The GAD-7 scale was applied to assess GAD symptom severity (Fig. 1). The GAD-2 scale was imposed as part of the GAD-7.

Total scores of the GAD-7 range between 0 and 21. Using a cut-off of 10 or greater, the GAD-7 yielded a sensitivity of 0.89 and a specificity of 0.82 in a large

FIGURE 1. The seven-item Generalized Anxiety Disorder scale (GAD-7)*.

GAD-7				
Over the last 2 weeks, how often have you been bothered by the following problems?	Not at all	Several days	More than half the days	Nearly every day
Feeling nervous, anxious or on edge	☐ 0	☐ 1	☐ 2	☐ 3
Not being able to stop or control worrying	☐ 0	☐ 1	☐ 2	☐ 3
Worrying too much about different things	☐ 0	☐ 1	☐ 2	☐ 3
Trouble relaxing	☐ 0	☐ 1	☐ 2	☐ 3
Being so restless that it is hard to sit still	☐ 0	☐ 1	☐ 2	☐ 3
Becoming easily annoyed or irritable	☐ 0	☐ 1	☐ 2	☐ 3
Feeling afraid as if something awful might happen	☐ 0	☐ 1	☐ 2	☐ 3
Total score ____ = Add columns ____ + ____ + ____				

* The GAD-2 subscale is composed of the first two items of the GAD-7 scale.

sample of primary care patients.¹⁴ Severity of GAD in primary care samples is represented by scores of ≥ 5 , ≥ 10 , and ≥ 15 for mild, moderate, and severe levels of anxiety symptoms, respectively.^{14,15} Total scores of the GAD-2 range between 0 and 7. A cut point of three or greater was recommended in two independent primary care studies to distinguish between GAD cases and non-cases.^{15,16}

The Patient Health Questionnaire (PHQ)-8 depression scale was used to assess depression symptoms and severity. The good psychometric properties of this questionnaire have been shown in several studies.²⁰

The Structured Clinical Interview for DSM-IV (SCID)²¹ was conducted via telephone with the participants of the study within 4 weeks after the home visits (mean time between completion of the questionnaire and interview: 25.5 days). The SCID interviews were conducted by a trained rater who was blinded in regard to the outcome of the GAD-7 scale. Current DSM-IV diagnoses as well as lifetime diagnoses of GAD, panic disorder, agoraphobia, specific phobia, social phobia, post-traumatic stress disorder, and depressive disorders were assessed. The SCID is a reliable and valid instrument to diagnose mental disorders.

Sample Size Calculation

Sample size calculation was estimated with respect to the sensitivity of the GAD-7 scale. Based on the results of Spitzer et al.,¹⁴ we expected a sensitivity of 90%. According to Flahault et al.,²² we needed 31 cases to ensure that the minimum lower limit of the 95% confidence interval (CI) would not be less than 65%. Assuming a prevalence of GAD in the elderly population of 7%,^{15,23} a total of 442 elderly persons was required for the study.

Statistical Analysis

Comparisons between excluded and included persons were done by using *t* tests and χ^2 tests for the variables gender, age, family status, and formal education. Point prevalence of GAD in the elderly population was estimated with its 95% CIs.

Descriptive statistics were provided for SCID diagnoses, GAD-7 and GAD-2 scores, and PHQ-8 scores. Pearson's correlation coefficient was calculated to measure the association between GAD severity and health-related quality of life or depression severity.

Internal consistency (i.e., the extent to which the items are correlated) was determined by calculating Cronbach's alpha.²⁴ A good internal consistency ($0.7 \leq \alpha \leq 0.9$) can be seen as a precondition of the fact that the summation of the single item scores to a total score is meaningful.²⁵

The SCID diagnosis of GAD constituted the criterion standard to determine the sensitivity and specificity of the GAD-7 and GAD-2 scales. We calculated operating characteristics (sensitivity, specificity) for various cut-off points of the GAD scales. We conducted receiver-operating characteristic curve analyses to determine the areas under the curve (AUCs) as a measure of diagnostic accuracy for the GAD-7 and GAD-2 scales. The AUC ranges between 0 and 1; its value reflects the probability that a randomly chosen case will score higher than a randomly chosen non-case.²⁶ Thus, an AUC value of 0.5 can be achieved by chance alone (no discrimination exists), whereas a test with perfect discrimination gives an AUC of 1.0.²⁷ According to Swets,²⁸ an AUC of 0.5–0.7 indicates low accuracy, 0.7–0.9 a moderate accuracy, and 0.9–1.0 indicates a high accuracy of the test. Statistical analysis was performed using PASW Statistics 18.0 for Windows and SAS, version 9.2.

TABLE 1. Baseline Characteristics of the Study Sample

Demographic Variable	Total (N = 438)		No Anxiety Disorder (N = 373)		GAD (N = 27)	
	N	%	N	%	N	%
Age, yr						
58–64	123	28.1	101	27.1	6	22.2
65–74	231	52.7	201	53.9	14	51.9
75–82	84	19.2	71	19.0	7	25.9
Gender						
Female	242	55.2	195	52.3	22	81.5
Male	196	44.8	178	47.7	5	18.5
Education, yr						
<9	7	1.6	5	1.3	1	3.7
9–10	359	82.0	311	83.6	19	70.4
11–12	40	9.1	31	8.3	4	14.8
>12	30	6.9	25	6.7	3	11.1
Marital status						
Never married	18	4.1	16	4.5		
Married	292	66.7	255	71.4	16	59.3
Divorced, widowed	109	24.9	86	24.1	10	37.0

RESULTS

Sample Characteristics

Included in the present study were 438 elderly persons. The demographic characteristics of the participants are shown in Table 1. Participants' age ranged between 58 and 82 years, with 52% of the participants aged 65 to 74; 55% were women.

Prevalence of GAD and Distribution of GAD-7 and GAD-2 Scores

In total, 27 participants met the DSM-IV criteria for current GAD according to the SCID (6.2%; 95% CI: 3.9%–8.2%). Fourteen of the 27 participants with GAD (51%) had, in addition, a diagnosis of lifetime major depression (of these 14 participants with GAD, two also had a current major depression diagnosis and four had an additional anxiety disorder). Another four persons with GAD had an additional anxiety disorder without comorbid depression. Thus, only nine participants had a single diagnosis of GAD without an additional diagnosis of anxiety or depression.

According to the Anatomical Therapeutic Chemical Classification System (ATC) of the nervous system, the following medication was taken by the 27 participants with GAD: 5 patients (19%) used

TABLE 2. Mean Scores and Standard Deviations for Each Item of the GAD-7 Stratified by the SCID Diagnosis GAD

GAD-7 Item	SCID diagnosis			
	No GAD (N = 411)		GAD (N = 27)	
	Mean	SD	Mean	SD
Feeling nervous, anxious, or on edge	0.39	0.61	1.30	0.87
Not being able to stop or control worrying	0.15	0.43	1.30	1.07
Worrying too much about different things	0.26	0.58	1.19	1.08
Trouble relaxing	0.22	0.54	1.07	1.00
Being so restless that it is hard to sit still	0.21	0.56	0.52	0.94
Becoming easily annoyed or irritable	0.34	0.59	0.74	1.02
Feeling afraid as if something awful might happen	0.11	0.37	0.56	0.80

Notes: SD: standard deviation.

antidepressant medication (ATC code N06A; 1 mirtazapine, 1 doxepin, 1 amitriptyline, 1 opipramol, 1 venlafaxine), 3 patients (11%) used benzodiazepines (ATC code N05BA; 2 oxazepam, 1 clobazam), 1 patient (4%) used tiapride (ATC code N05AL03), and 1 patient (4%) used carbamazepine (ATC code N03AF01). Antidepressants and benzodiazepines were combined in two cases (1 mirtazapine and oxazepam, 1 amitriptyline and oxazepam). Altogether, 8 of 27 GAD patients (30%) were treated with at least one medication for the nervous system.

Table 2 provides the mean scores for each of the seven items of the GAD-7 and shows that elderly persons with a GAD diagnosis scored highest on the first and second item of the GAD-7 (feeling nervous, anxious, or on edge and unable to stop worrying). Table 3 shows the mean scores and standard deviations of the GAD-7 and GAD-2 scales and the PHQ-8 in the total sample as well as in participants with various diagnoses.

The mean GAD-7 and GAD-2 scores of the total sample were markedly lower than the mean GAD-7 and GAD-2 scores of patients with GAD. Patients with a current depressive disorder showed the highest mean GAD-7 and GAD-2 scores. A similar pattern was found regarding depression severity: Patients with a GAD or current depressive disorder had the highest PHQ-8 scores.

Operating Characteristics of the GAD-7 and the GAD-2

Table 4 shows the operating characteristics of the two GAD scales at various cut points. The sensitivity strongly and noticeably decreases with the increase of the cut points. The reason for this fast decrease is the surprising finding that 10 elderly persons with a GAD diagnosis scored lower than five in the GAD-7 scale. Thus, at a cut point of three or greater, the sensitivity of the GAD-7 scale is high (0.89). However, specificity at a cut point of three or greater is quite low (0.76). At a cut point of five or greater, sensitivity of the GAD-7 is 0.63 and specificity is 0.9. Thus, a cut point of five or greater appears to be the optimal cut point for detecting GAD with the GAD-7 scale. Correspondingly, with a sensitivity of 0.67 and a specificity of 0.90, a cut point of two or greater appears to be the optimal cut point for detecting GAD with the GAD-2 scale.

Figure 2 shows the receiver-operating characteristic curves of the GAD-7 and the GAD-2, illustrating sensitivity and specificity at various cut points. The AUC of the GAD-7 is 0.88 (95% CI: 0.83–0.93); the AUC of the GAD-2 is 0.87 (95% CI: 0.80–0.94). These AUCs indicate moderate accuracy of the questionnaires regarding the discrimination between persons with GAD and those without GAD.

Internal Consistency and Construct Validity

Cronbach's alpha for the GAD-7 and the GAD-2 were 0.82 and 0.71, respectively, indicating good internal consistency for both scales in this study sample. Correlations between GAD-7 sum scores and health related quality of life showed that higher GAD-7 scores were related to decreased PCS and MCS. There was a strong correlation between GAD-7 scores and MCS ($r = -0.48$, $p < 0.0001$) and a moderate but still significant correlation between GAD-7 sum scores and PCS ($r = -0.20$, $p < 0.0001$). Correspondingly, the correlation between the GAD-2 and the MCS ($r = -0.44$, $p < 0.0001$) and between the GAD-2 and the PCS ($r = -0.22$, $p < 0.0001$) was substantial.

The correlation between the GAD scores and PHQ-8 depression scores was high (GAD-7: $r = 0.70$, $p < 0.0001$; GAD-2: $r = 0.68$, $p < 0.0001$); this is consistent with previous studies. Associations between various demographic variables and anxiety

TABLE 3. Mean and Standard Deviations of the GAD-7, the GAD-2, and PHQ-8 Scales Stratified According to SCID Diagnoses

Diagnosis According to SCID	Total (N = 438)	GAD (N = 27)	Depressive Disorder (Current) (n = 10)	Depressive Disorder (Lifetime) (n = 68)	Anxiety Disorder (N = 65)	Two or More Mental Disorders (n = 13)
GAD-7						
Mean	2.00	6.67	7.8	3.72	5.04	6.96
SD	2.88	4.36	6.14	4.17	4.32	5.14
GAD-2						
Mean	0.66	2.59	2.9	1.31	1.85	2.31
SD	1.08	1.69	1.60	1.54	1.66	1.75
PHQ-8						
Mean	2.73	8.18	9.3	4.87	6.07	7.77
SD	3.28	4.45	5.62	4.41	4.53	3.83

Notes: SD: standard deviation.

TABLE 4. Operating Characteristics of the GAD-7 and GAD-2 at Various Cut Points

Cut Point	GAD-7		GAD-2	
	Sensitivity [95%-CI]	Specificity [95%-CI]	Sensitivity [95%-CI]	Specificity [95%-CI]
≥1	1.00 [0.87; 1.00]	0.42 [0.37; 0.47]	0.93 [0.76; 0.99]	0.64 [0.59; 0.68]
≥2	0.93 [0.75; 0.99]	0.63 [0.58; 0.67]	0.67 [0.46; 0.83]	0.90 [0.86; 0.93]
≥3	0.89 [0.70; 0.98]	0.76 [0.71; 0.79]	0.44 [0.25; 0.65]	0.96 [0.93; 0.97]
≥4	0.63 [0.42; 0.81]	0.84 [0.80; 0.88]	0.37 [0.19; 0.58]	0.98 [0.96; 0.99]
≥5	0.63 [0.42; 0.81]	0.90 [0.87; 0.93]	0.15 [0.04; 0.34]	0.99 [0.98; 1.00]
≥6	0.52 [0.32; 0.71]	0.94 [0.91; 0.96]	0.04 [0.01; 0.19]	1.00 [0.99; 1.00]
≥7	0.44 [0.25; 0.65]	0.95 [0.93; 0.97]	n/a	n/a
≥8	0.37 [0.19; 0.58]	0.97 [0.94; 0.98]	n/a	n/a
≥9	0.37 [0.19; 0.58]	0.97 [0.95; 0.98]	n/a	n/a
≥10	0.26 [0.11; 0.46]	0.98 [0.96; 0.99]	n/a	n/a
≥11	0.22 [0.08; 0.42]	0.98 [0.97; 1.00]	n/a	n/a
≥12	0.11 [0.02; 0.29]	0.99 [0.98; 1.00]	n/a	n/a
≥13	0.07 [0.00; 0.24]	0.99 [0.98; 1.00]	n/a	n/a
≥14	0.04 [0.00; 0.19]	0.99 [0.98; 1.00]	n/a	n/a

severity according to the GAD-7 and GAD-2 scales are shown in Table 5.

Women had significantly higher GAD-7 and GAD-2 scores as compared with men. Persons ages 65 to 74 had the lowest GAD-7 and GAD-2 scores; the difference in GAD scores between age groups was significant (Table 5).

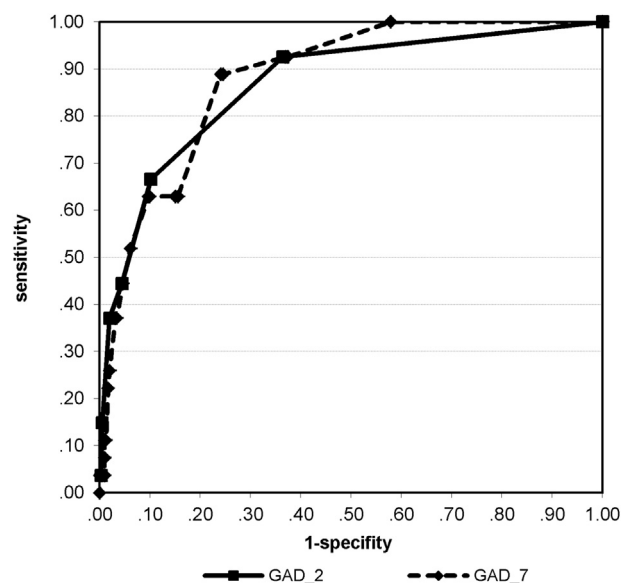
DISCUSSION

This is the first study that has investigated the validity of the GAD-7 and the ultrashort GAD-2 in a population-based sample of elderly people ages 58 to 82. Results indicate that in elderly people the cut points for detecting GAD should be lowered from 10 to 5 for the GAD-7 and from 3 to 2 for the GAD-2. Using these cut points yields a high specificity (0.9 for GAD-7 and GAD-2) and a reasonable sensitivity (0.63 for GAD-7 and 0.67 for GAD-2) for detecting a GAD.

What could be the reason for this markedly lower threshold for detecting GAD in elderly people as compared with the thresholds determined in previous studies?^{14–16} First, in this study, we investigated a population-based sample and not a clinical sample. It is probable that persons randomly chosen from the general population are less inclined to admit to excessive and uncontrollable worries or to problems of concentration or irritability in a questionnaire.

Second, the blinded interviewer reported that several participants diagnosed with GAD at first negated the screening question for GAD and refused to speak about their uncontrollable worries; a second inquiry, however, led to the affirmation of the SCID screening question for GAD, “Over the past six months, did you feel particularly anxious, nervous, or on edge?” This information supports our initial assumption that the threshold for detecting GAD could be lower for elderly people because they tend

FIGURE 2. The receiver-operating characteristic (ROC) curves for various cut points of the GAD-7 and the GAD-2.



to complain less about psychological burdens than younger people.⁹ We conjectured that elderly people may be more embarrassed to report uncontrollable worries or worries over minor matters than younger people.

Third, in some cases, the presence of a study doctor while answering the GAD-7 could have led to a response tendency toward social desirability, thus negating existing symptoms of worrying too much and feeling afraid. Several studies support the hypothesis that an age-related increase in social desirability may alter the patterns in self-reported data; thus, some of the older people tend toward the suppression of physical and psychological impairment in scientific or clinical surveys.²⁹ However, this response behavior could also be correlated to age-related coping strategies to manage both mental and physical limitations.³⁰

Based on our data, we cannot determine with certainty what the underlying reason for the lower threshold is. However, the inference of our study is that using the GAD-7 or the GAD-2 as screening instruments in population-based samples of elderly people, the threshold for detecting GAD should be lowered.

Using cut points of five or greater for the GAD-7 and two or greater for the GAD-2 yield a lower

sensitivity than originally targeted in our sample size calculation. However, these cut points appear to be optimal cut points because they combine a reasonable sensitivity with a high specificity of 0.9. With a 6.2% prevalence of GAD in our sample and similar or lower estimates in other studies, the high specificity ensures that most elderly persons not suffering from GAD are properly classified as non-cases (which corresponds to a low false-positive rate). However, we have to admit that at the cut points of five (GAD-7) and two (GAD-2), the sensitivities of 0.63 and 0.67, respectively, are lower than we would have expected.

How does the GAD-2 compare with its full version, the GAD-7? It should be emphasized that the GAD-2 gives clinical information only about the two cardinal symptoms of GAD, namely nervousness and inability to stop worrying (Fig. 1). In contrast, the GAD-7 asks for most diagnostic features of GAD, including trouble relaxing, restlessness, and excitability. Thus, the GAD-7 provides more clinical information than the GAD-2; the GAD-7 also allows for the opportunity to check more diagnostic features of GAD that relate to DSM-IV criteria. A direct comparison of the diagnostic accuracies using the AUCs of GAD-7 and GAD-2 (0.88 and 0.87, respectively) shows nearly identical results. Of note, at the recommended cut points for the diagnosis of GAD, the sensitivity of the GAD-2 was somewhat higher compared with the sensitivity of the GAD-7. However, thus far no study has compared sensitivity in the changes of the GAD-7 and the GAD-2 that prevents one from evaluating whether one GAD scale is better than the other. In summary, the GAD-7 appears to be the more comprehensive measure as compared with the GAD-2. Nevertheless, for clinical applications or research studies where brevity is desirable, the GAD-2 is certainly an appropriate alternative.

An important result of the study is that we could confirm the expected association between GAD severity and health related quality of life. Increased GAD scale scores are strongly associated with decreased mental quality of life and are also significantly related to lower scores in physical quality of life. The high correlation between GAD scale scores and severity of depression symptoms reflects the well-known fact that GAD is highly comorbid with depression. In addition, the GAD scale scores in

TABLE 5. GAD-7 and GAD-2 Scores of the Study Participants, Stratified by Demographic Variables (N = 438)

	GAD-7 Score, M (SD)	t / F Value	p Value GAD-7	GAD-2 Score M (SD)	t / F Value	p Value GAD-2
Gender						
Female	2.28 (3.35)	$t_{(414.6)} = 2.35$	0.02	0.81 (1.25)	$t_{(413.4)} = 3.30$	0.001
Male	1.66 (2.13)			0.48 (0.79)		
Age, yr						
55–64	2.20 (3.28)	$F_{(2,435)} = 3.21$	0.04	0.67 (1.06)	$F_{(2,435)} = 4.96$	0.007
65–74	1.69 (2.29)			0.55 (0.98)		
75–80	2.56 (3.57)			0.98 (1.33)		
Marital status						
Never married	1.39 (1.85)	$F_{(2,416)} = 0.40$	0.67	0.44 (0.78)	$F_{(2,416)} = 3.02$	0.05
Married	1.94 (2.83)			0.57 (0.97)		
Divorced, widowed	2.04 (2.95)			0.84 (1.29)		
Education, yr						
<9	2.57 (4.28)	$F_{(3,432)} = 0.88$	0.45	0.86 (2.27)	$F_{(3,432)} = 1.10$	0.35
9–10	1.92 (2.70)			0.64 (1.01)		
11–12	2.60 (3.71)			0.90 (1.35)		
>12	1.72 (2.60)			0.47 (1.04)		

Notes: p values are associated with t tests or analyses of variance for differences in mean values between different categories of the demographic variables. SD: standard deviation.

elderly women were significantly higher compared with elderly men. These various correlations demonstrate the construct validity of both the GAD-7 and the GAD-2 in elderly people.

According to the SCID, we found a prevalence rate of GAD of 6.2% that is comparable with the prevalence estimations of other studies in elderly people.^{1,31} However, several studies in older adults found a considerably lower GAD prevalence of 2.8%.^{2,6} We cannot explain these differences in prevalence estimations based on various studies; presumably they are associated with cultural differences as well as with methodologic issues.

Consistent with the results of previous studies, we found that a GAD diagnosis is highly related to a comorbid diagnosis of depression.³² This strong association between GAD and depressive symptoms makes it difficult to form a diagnosis.³³ The great overlap between GAD and other mental disorders gives rise to the controversy regarding GAD as a diagnostic entity.¹ Some experts argue that GAD and depression are not separate disorders but rather represent one disorder at a different stage of development.³⁴ However, Lenze et al.³⁵ indicated that GAD in elderly people was typically chronic and could be separated from major depression; they reported that GAD in late life tended to persist even if comorbid depression remitted. In addition, Beekman et al.³⁶ showed that risk factors for anxiety disorders in late life differed from risk factors for major

depression. Thus, there is evidence that GAD and depression are distinct disorders with a considerable phased overlap. An important recommendation of several experts for the fifth edition of the DSM is, therefore, that the “DSM-5 should provide guidelines on the assessment of comorbid anxiety and depression in older adults.”⁹

The study has several limitations. First, it is possible that the study sample is not representative of the general German population ages 55 to 80. In the 8-year follow-up of the ESTHER study, all ESTHER participants were asked if they wanted to participate in a home visit with a study doctor. It is possible that persons who agreed to participate in the home visits were healthier or less impaired regarding psychosocial aspects of life as compared to persons who declined to participate. However, the percentage of 6.2% of GAD diagnoses found in our study sample indicates that the sample is comparable with other population-based studies investigating anxiety disorders in elderly people. Second, the GAD-7 questionnaire was completed on average 25 days before the diagnostic interview was conducted. We cannot rule out that the time interval between self-report and expert assessment influenced the agreement (or disagreement) between the two measures. Third, the presence of a study doctor could have biased the completion of the GAD-7 questionnaire. It is possible that some elderly persons tended to score lower with respect to

specific symptoms of GAD because a study doctor was present. However, this interpretation remains speculative. It is also possible that participants would feel more comfortable disclosing psychological symptoms to physicians.

CONCLUSION

This is the first study that has investigated the validity of the GAD-7 and GAD-2 scales in elderly people. The criterion validity of both the GAD-7 and the GAD-2 appears to be reasonable in the general elderly population aged 58 to 82. Results indicate that the cut-off score of the GAD-7 for detecting GAD in elderly people should be lowered from 10 to 5, and that the cut-off score of the GAD-2 should be lowered from 3 to 2. The study confirms that in elderly people the overlap between GAD and other anxiety disorders, as well as between GAD and depression, is

large. It is obvious that GAD cannot be separated from other anxiety disorders or from depression by the use of a simple questionnaire. The study results establish a similar validity of the GAD-7 scale and the GAD-2 scale. It is therefore recommended that both scales be used as screening instruments for GAD. More refined clinical diagnostics are still necessary to assess the possible occurrence of an anxiety–depression syndrome in elderly people with high GAD-7 scores; in addition, it can provide a solid foundation for an adequate treatment plan.

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